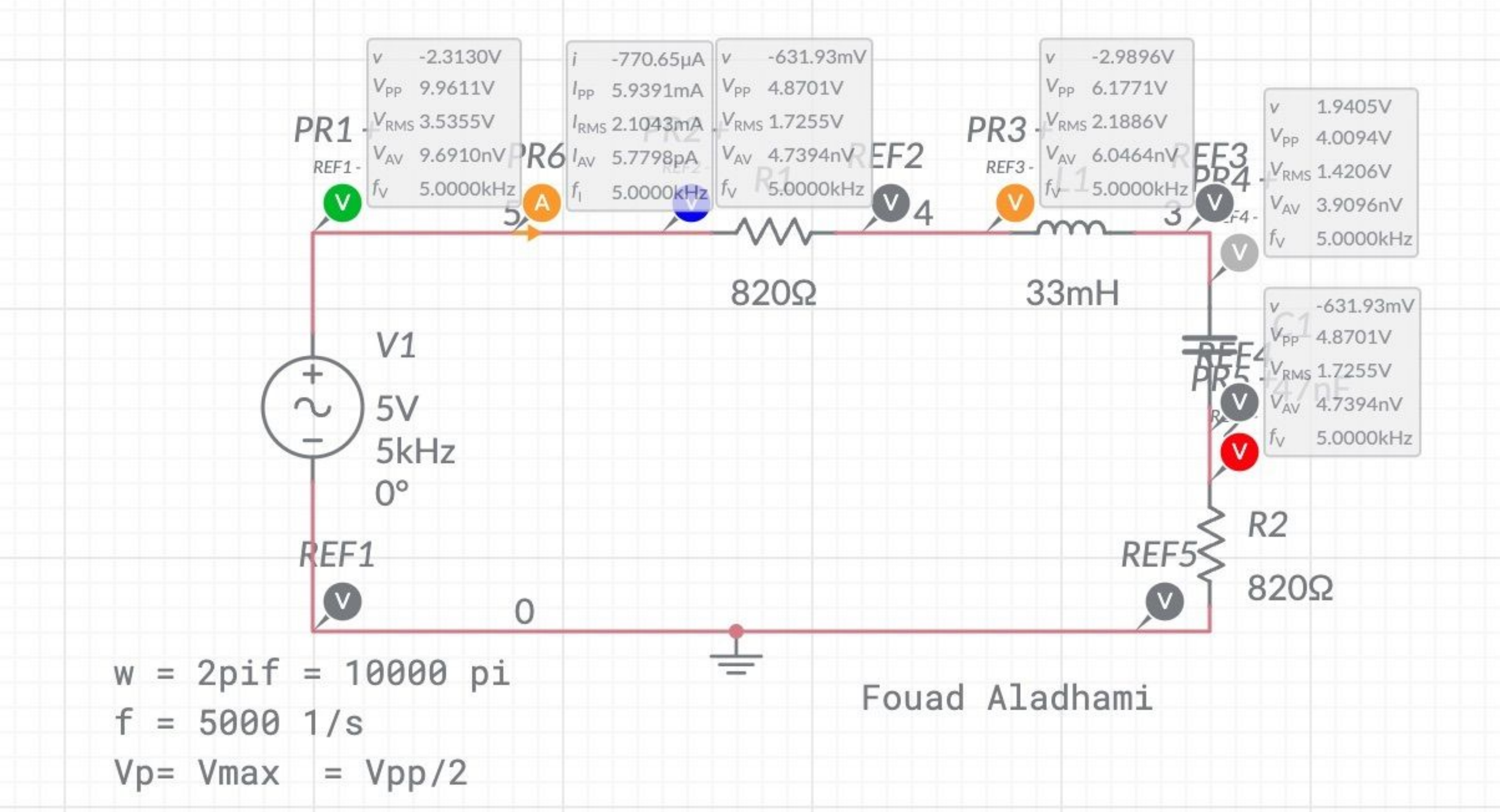

v -2.3130V
VPP 9.9611V
VRMS 3.5355V
VAV 9.6910nV
fV 5.0000kHz
i -770.65µA
IPP 5.9391mA
IRMS 2.1043mA
IAV 5.7798pA
fI 5.0000kHz
v -631.93mV
VPP 4.8701V
VRMS 1.7255V
VAV 4.7394nV
fV 5.0000kHz
v -2.9896V
VPP 6.1771V
VRMS 2.1886V
VAV 6.0464nV
fV 5.0000kHz
v 1.9405V
VPP 4.0094V
VRMS 1.4206V
VAV 3.9096nV
fV 5.0000kHz
v -631.93mV
VPP 4.8701V
VRMS 1.7255V
VAV 4.7394nV
fV 5.0000kHz
PR1
REF1
PR6
REF2
PR3
REF3
PR4
REF4
PR5
REF5
820Ω
33mH
V1
5V
5kHz
0°
R2
820Ω
0
w = 2pif = 10000 pi
f = 5000 1/s
Vp= Vmax = Vpp/2
Fouad Aladhami

Interactive 1

Stopped | None

C1 C2

PR1: V(5)-V(0)
PR5: V(2)-V(0)
PR2: V(5)-V(4)
PR3: V(4)-V(3)
PR4: V(3)-V(2)
PR6: I(R1)

Voltage (V): 6, 5, 4, 3, 2, 1, 0, -1, -2, -3, -4, -5, -6

Current (A): 3m, 2m, 1m, 0, -1m, -2m, -3m

2.21665 2.2167 2.21675 2.2168 2.21685 2.2169 2.21695 2.217

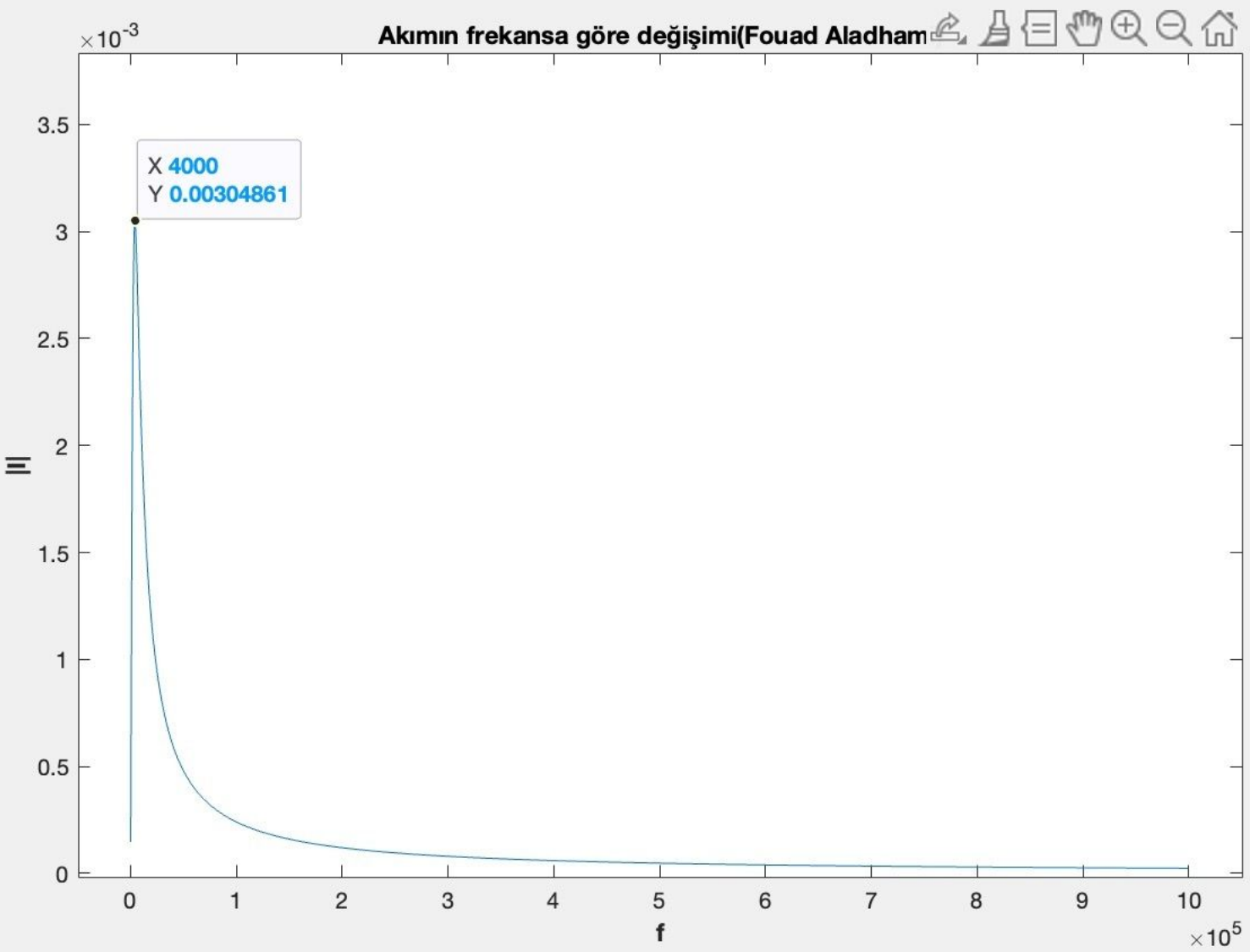
Time (s)

Cursor 1	Cursor 2	ΔX	1/ΔX
2.2167 s, 4.9782 V	2.2167 s, 2.0028 V	57.185 µs	17.487 kHz

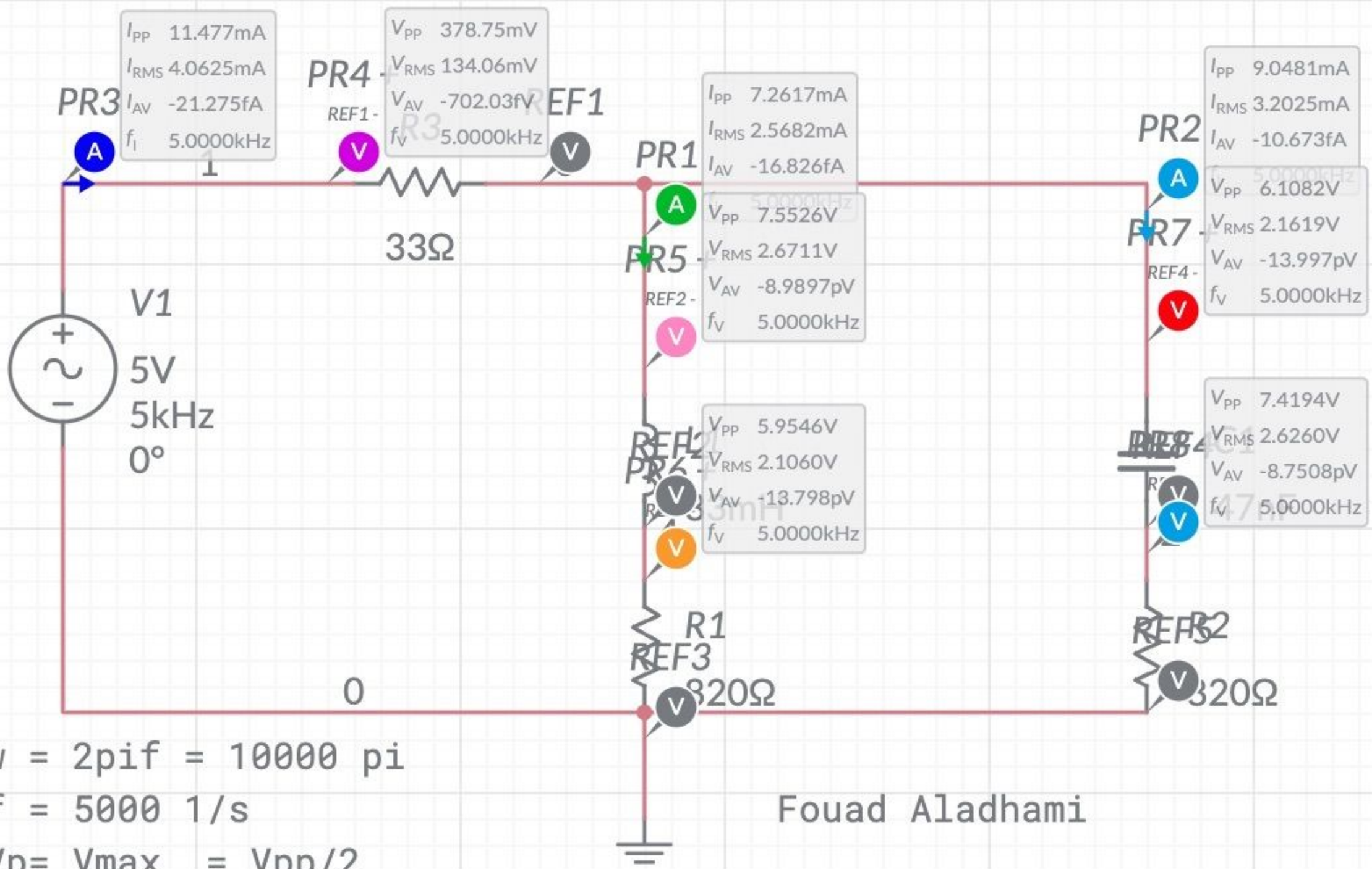
Editor - /Users/fuad/Library/CloudStorage/GoogleDrive-fouadiadhami@gmail.com/Drive'ım/Elektronik mühend...

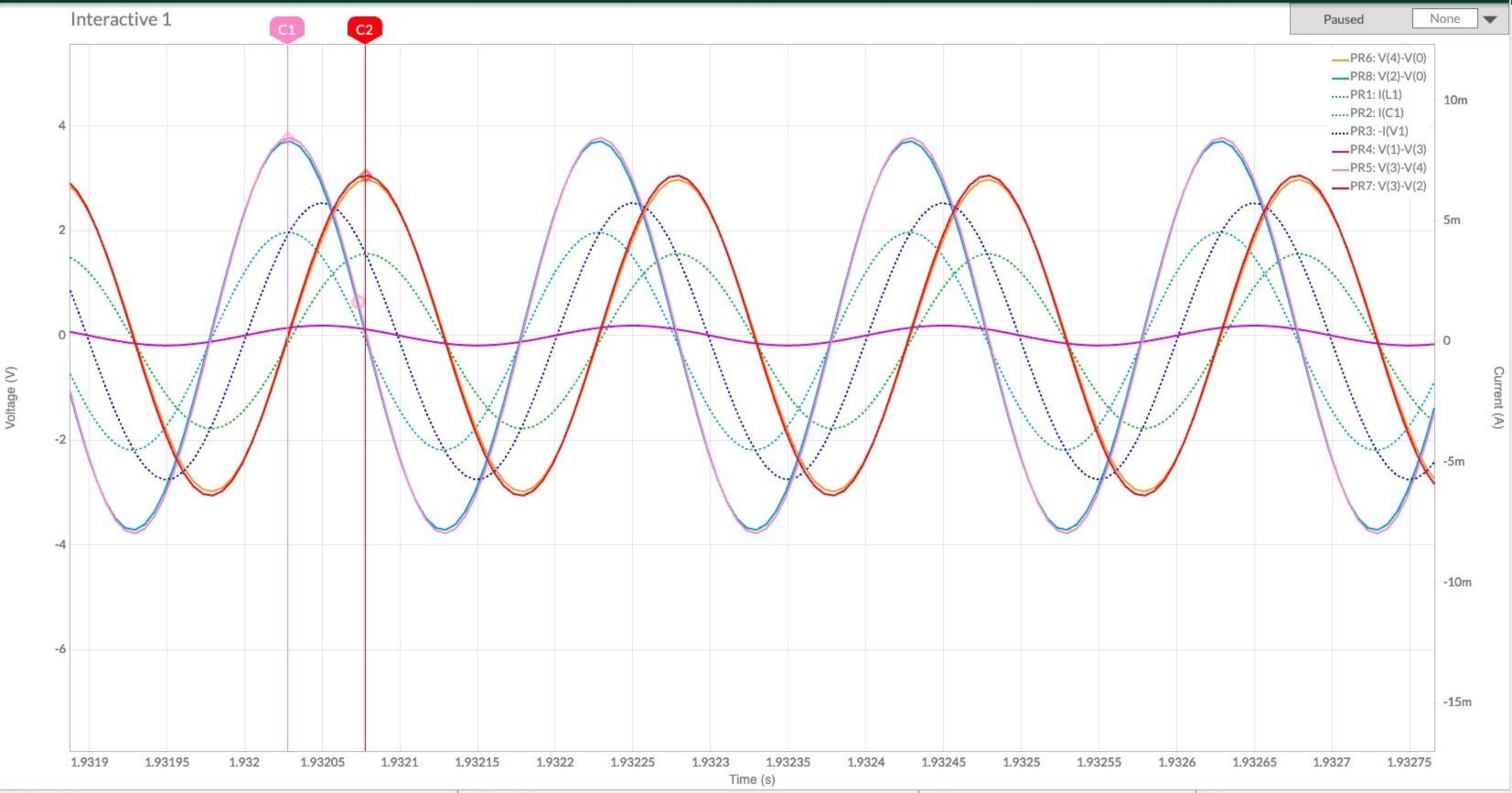
+5 | lab2.m | project2_1.m | project2_4.m | project2_2.m | project2_3.m | isprop.m | lab3.m

```
Vs = 5;
%w = 10^4 * pi;
f = 100:100:10^6;
w = 2.*pi.*f;
Z1 = 820 + 1i.*w.*33*10^-3;
Z2 = 820 + 1./(1i.*w.*47.*10^-9);
Zr1 = 820;
Zr2 = 820;
Zl = 1i *w*33*10^-3;
Zc = 1./(1i.*w.*47.*10^-9);
Vz1 = (Z1./(Z1+Z2)) .* Vs;
Vz2 = (Z2./(Z1+Z2)) .* Vs;
Vr1 = (Zr1./(Z1+Z2)) .* Vs;
Vr2 = (Zr2./(Z1+Z2)) .* Vs;
Vl = (Zl./(Z1+Z2)) .* Vs;
Vc = (Zc./(Z1+Z2)) .* Vs;
Is = Vr2./Zr2;
plot(f, abs(Is))
title("Akımın frekansa göre değişimi")
xlabel("f","FontWeight","bold")
ylabel("|I|","FontWeight","bold")
%Fouad Aladhami
```




```
%Fouad Aladhami
```


PR3
I_{PP} 11.477mA
I_{RMS} 4.0625mA
I_{AV} -21.275fA
f_I 5.0000kHz
PR4
V_{PP} 378.75mV
V_{RMS} 134.06mV
V_{AV} -702.03fV
f_V 5.0000kHz
REF1
33Ω
PR1
I_{PP} 7.2617mA
I_{RMS} 2.5682mA
I_{AV} -16.826fA
PR5
V_{PP} 7.5526V
V_{RMS} 2.6711V
V_{AV} -8.9897pV
f_V 5.0000kHz
REF2
PR2
I_{PP} 9.0481mA
I_{RMS} 3.2025mA
I_{AV} -10.673fA
PR7
V_{PP} 6.1082V
V_{RMS} 2.1619V
V_{AV} -13.997pV
f_V 5.0000kHz
REF4
V1
5V
5kHz
0°
V_{PP} 5.9546V
V_{RMS} 2.1060V
V_{AV} -13.798pV
f_V 5.0000kHz
V_{PP} 7.4194V
V_{RMS} 2.6260V
V_{AV} -8.7508pV
f_V 5.0000kHz
R1
REF3
320Ω
R2
320Ω
0
w = 2pif = 10000 pi
f = 5000 1/s
Vp= Vmax = Vpp/2
Fouad Aladhami



Cursor 1	Cursor 2	ΔX	1/ΔX
1.9320 s, 3.7638 V	1.9321 s, 3.0471 V	50.049 µs	19.980 kHz

```
Vr1phase = angle(Vr1).*180./pi;
Vr2 = I2.*Zr2;
Vr2abs = abs(Vr2);
Vr2phase = angle(Vr2).*180./pi;
Vl1 = I1.*Zl;
Vl1abs = abs(Vl1);
Vl1phase = angle(Vl1).*180./pi;
Vc1 = I2.*Zc;
Vc1abs = abs(Vc1);
Vc1phase = angle(Vc1).*180./pi;

subplot(2,1,1)
plot(f, abs(I1))
title("I1 Akımının frekansa göre değişimi(Fouad Aladhami)")
xlabel("f","FontWeight","bold")
ylabel("|I1|","FontWeight","bold")

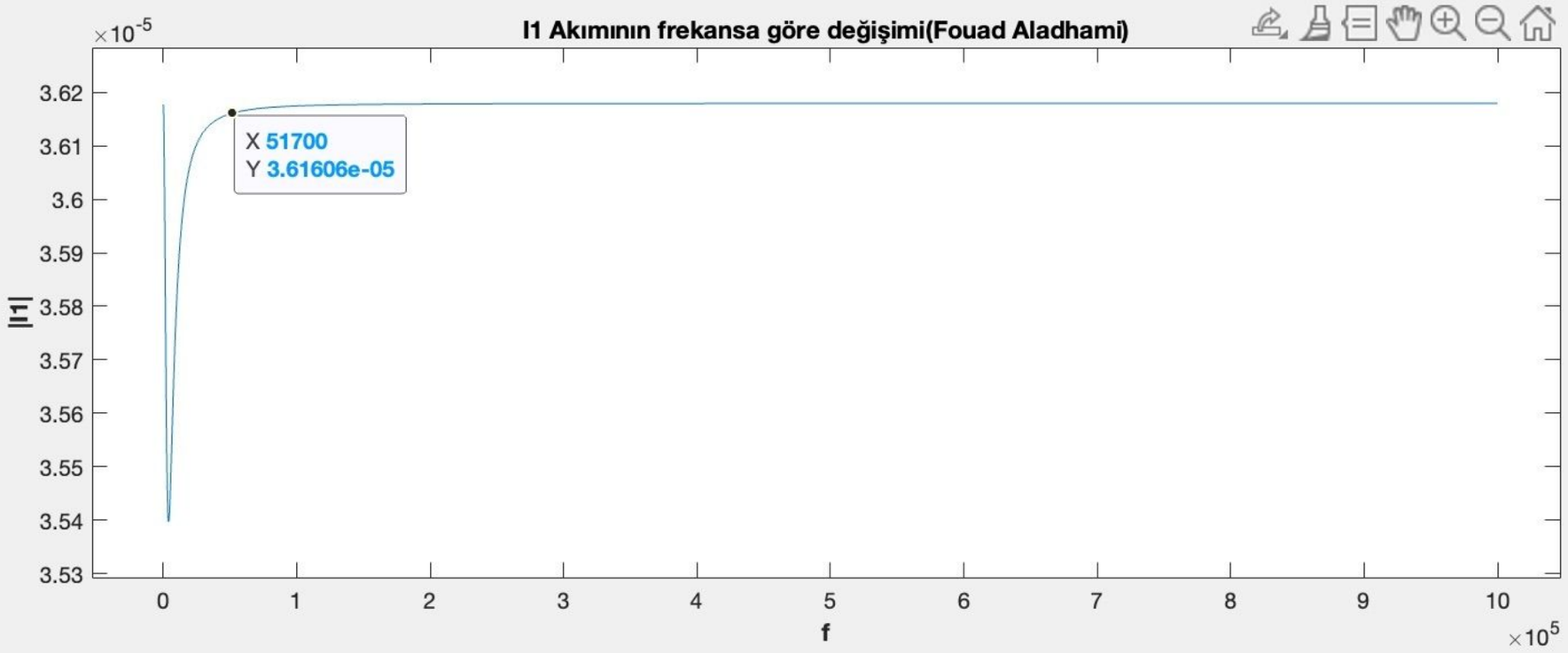
subplot(2,1,2)
plot(f,abs(I2))
title("I2 Akımının frekansa göre değişimi(Fouad Aladhami)")
xlabel("f","FontWeight","bold")
ylabel("|I2|","FontWeight","bold")
%Fouad Aladhami
```

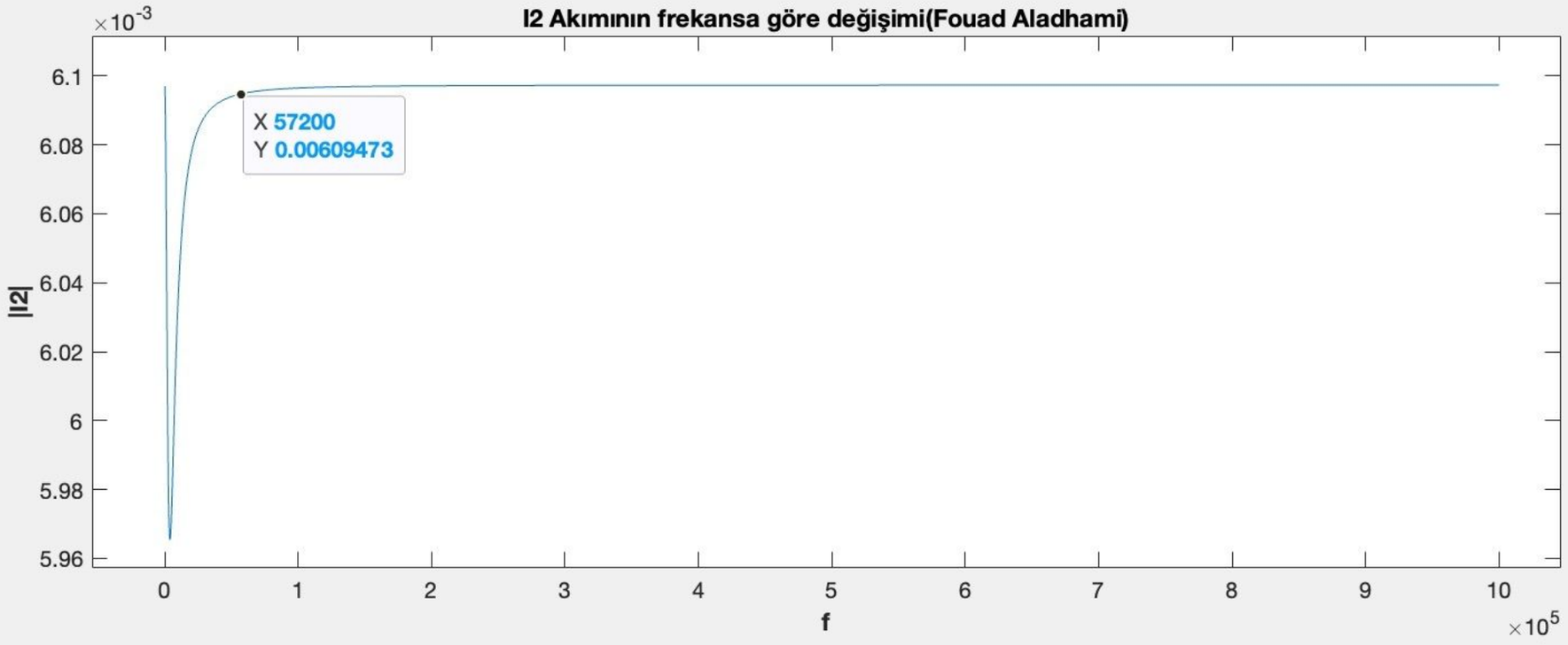
Editor - lab3_2.m

Variables - I1

project2_1.m | project2_4.m | project2_2.m | project2_3.m | isprop.m | lab3_1.m | lab3_2.m

```
Vs = 5;
%w = 10^4 * pi;
f = 100:100:10^6;
w = 2.*pi.*f;
Zr1 = 820;
Zr2 = 820;
Zl = 1i *w*33*10^-3;
Zc = 1./(1i.*w.*47.*10^-9);
Z1 = Zr1 + Zl;
Z2 = 820 + Zc;
Zeq = (Z1.*Z2) ./(Z1+Z2);
Is = Vs./Zeq;
Isabs = abs(Is)*1000;%ma
Isph = angle(Is).*180./pi;
I1 = (Z2/(Z1+Z2)).*Is;
Is1abs = abs(I1).*1000;%ma
Is1ph = angle(I1).*180./pi;
I2 = (Z1/(Z1+Z2)).*Is;
Is2abs = abs(I2).*1000;%ma
Is2ph = angle(I2).*180./pi;
Vr1 = I1.*Zr1;
Vr1abs = abs(Vr1);
Vr1phase = angle(Vr1).*180./pi;
```


I1 Akımının frekansa göre değişimi(Fouad Aladhami)
×10^-5
3.62
3.61
3.6
3.59
3.58
3.57
3.56
3.55
3.54
3.53
|I1|
X 51700
Y 3.61606e-05
0
1
2
3
4
5
6
7
8
9
10
f
×10^5


I2 Akımının frekansa göre değişimi(Fouad Aladhami)
×10^-3
6.1
6.08
6.06
6.04
6.02
6
5.98
5.96
|I2|
X 57200
Y 0.00609473
0
1
2
3
4
5
6
7
8
9
10
f
×10^5